AI-Assisted Coding

Assignment-6.3

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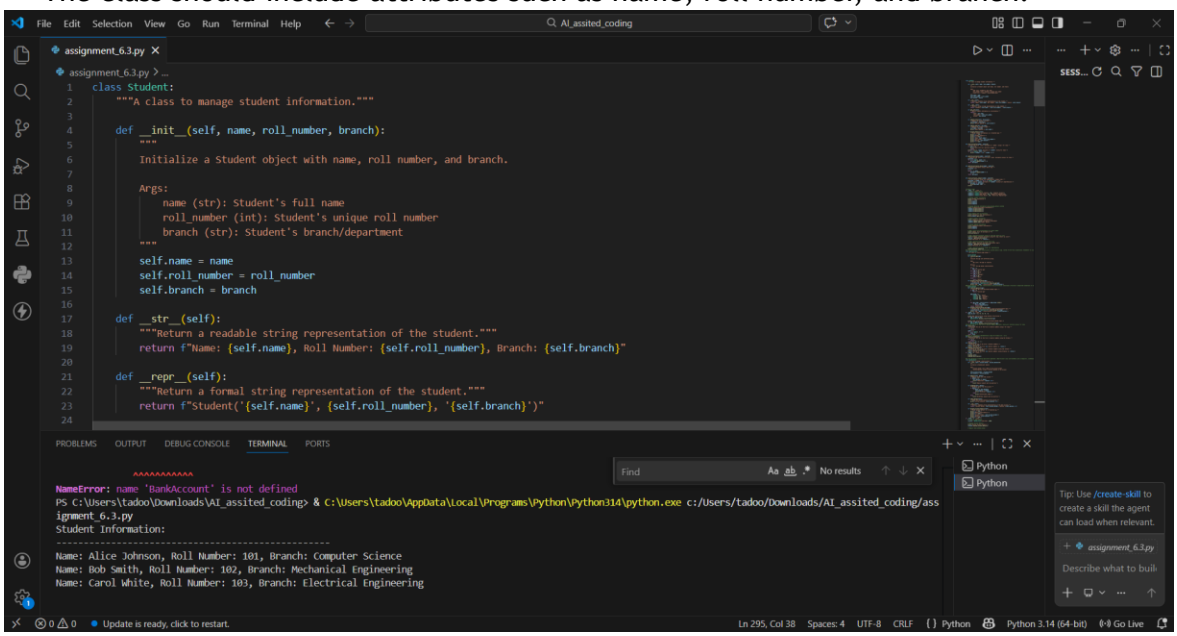
Batch: 45

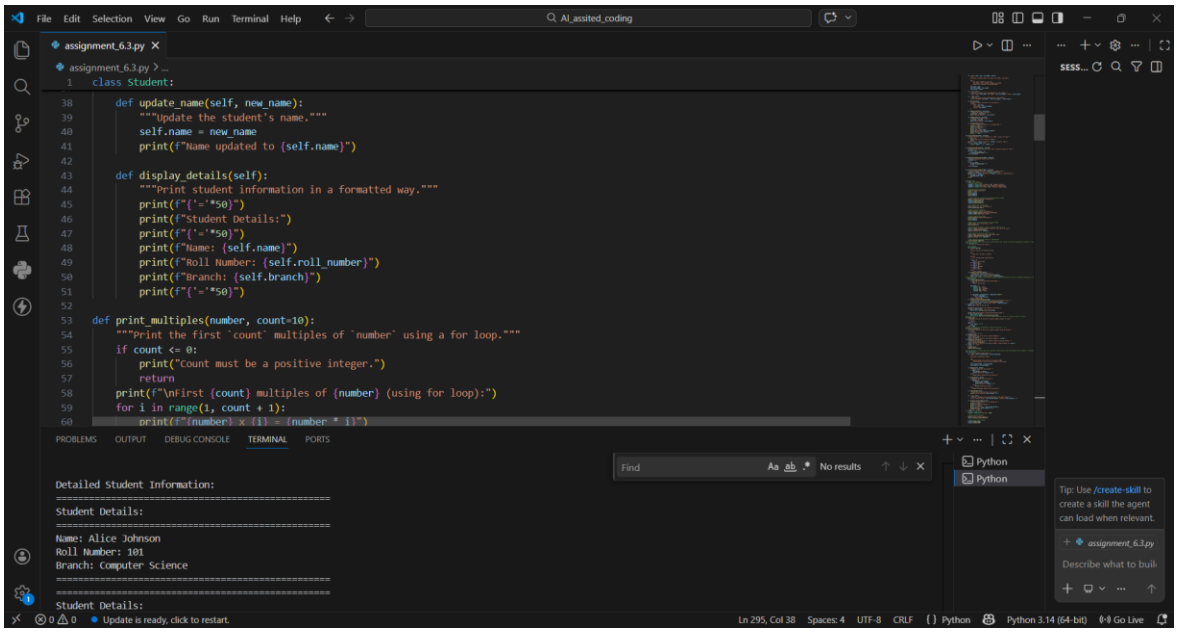
Task1: Classes (Student Class)

=>Use an AI tool (GitHub Copilot) to complete a Student class.

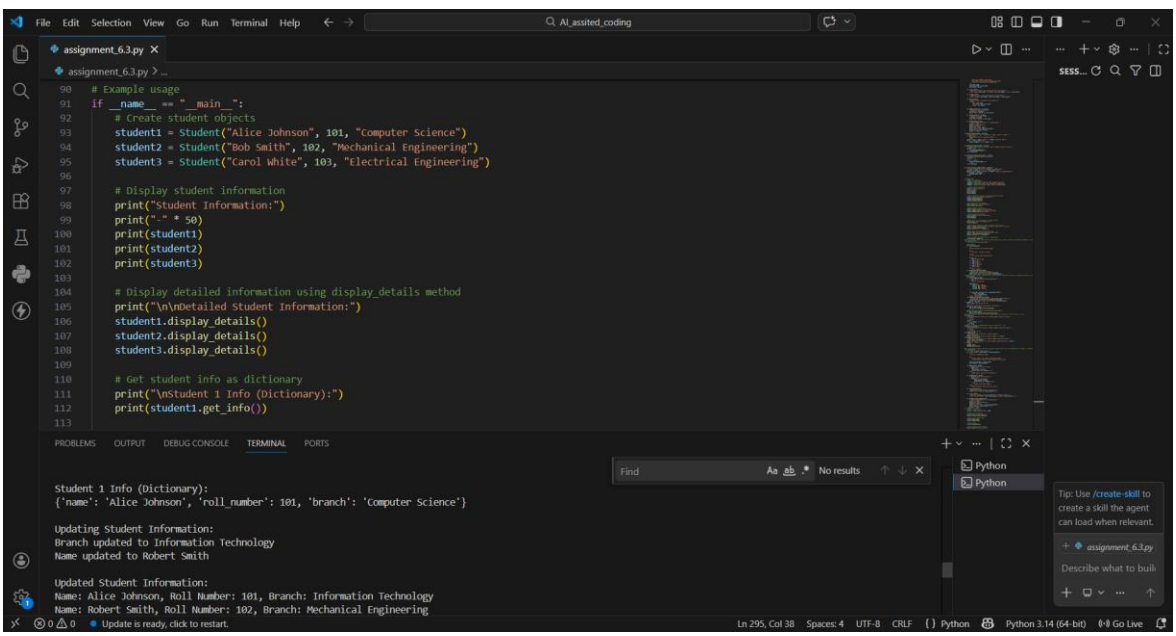
=>The class should include attributes such as name, roll number, and branch.



=>Add a method display_details() to print student information.



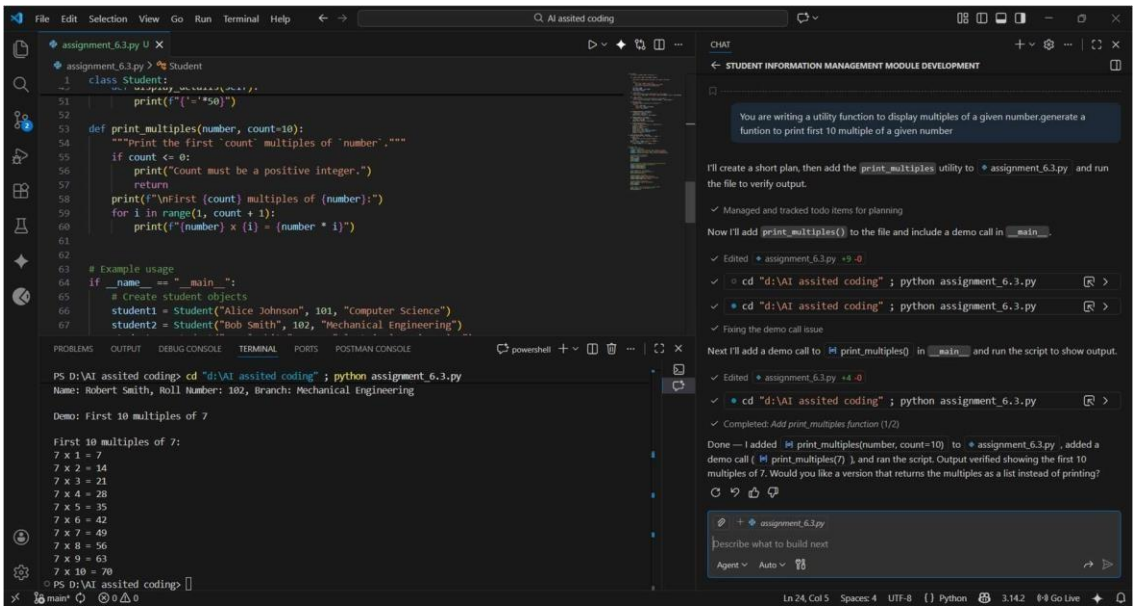
=>Execute the code and verify the output.



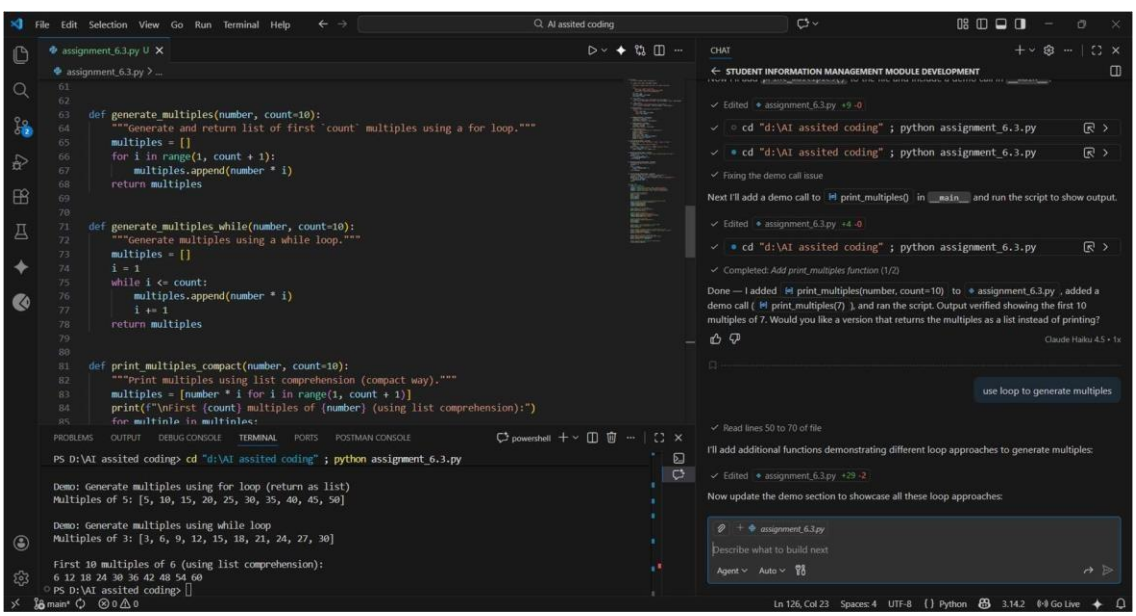
=>Analyze the code generated by the AI tool for correctness and clarity.

Task Description #2: Loops (Multiples of a Number)

=>generate a funtion to print first 10 multiple of a given number

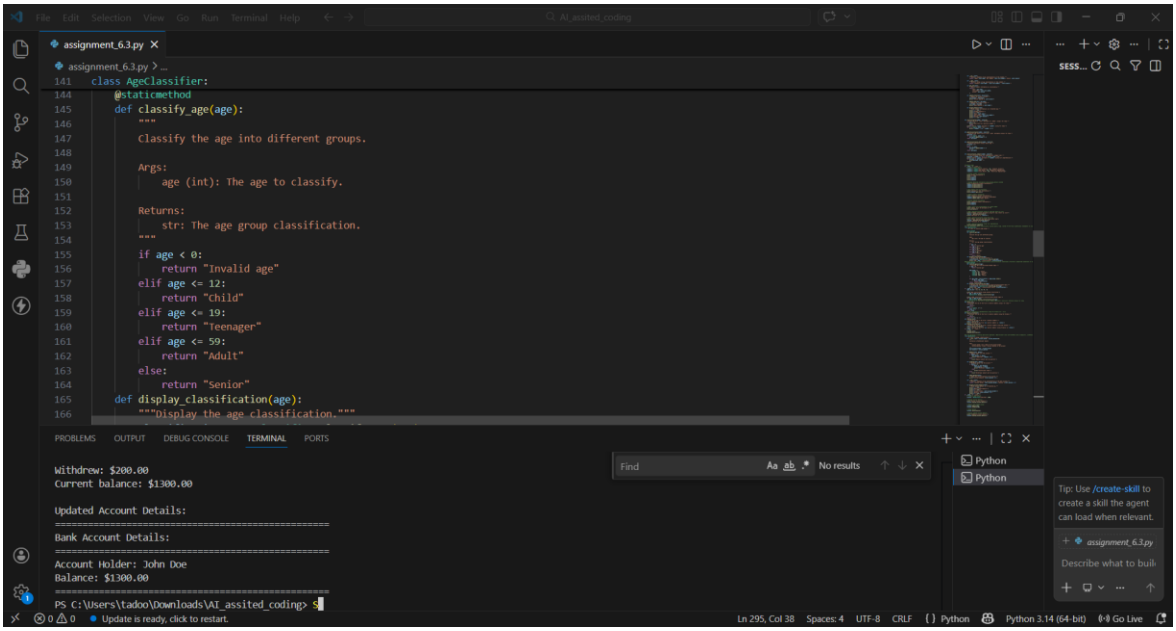


=>Ask the AI to generate the same functionality using another controlled looping structure (e.g.,while instead of for).

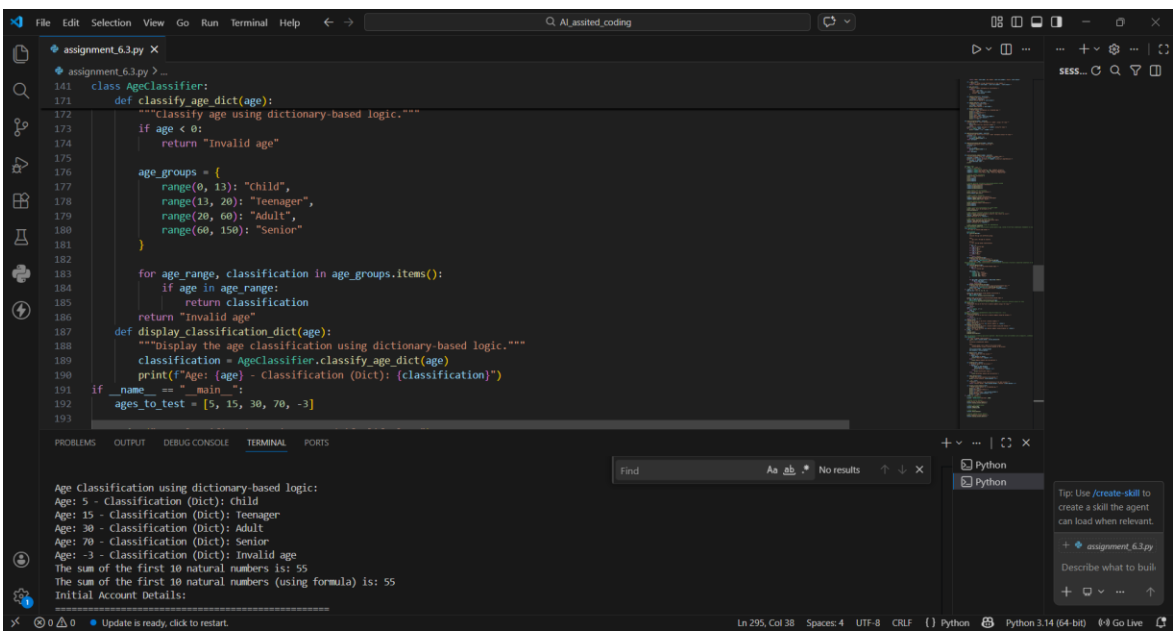


Task Description #3: Conditional Statements (Age Classification)

You are building a basic classification system based on age. nested if-elif-else conditional statements to classify age groups(e.g., child, teenager, adult, senior).

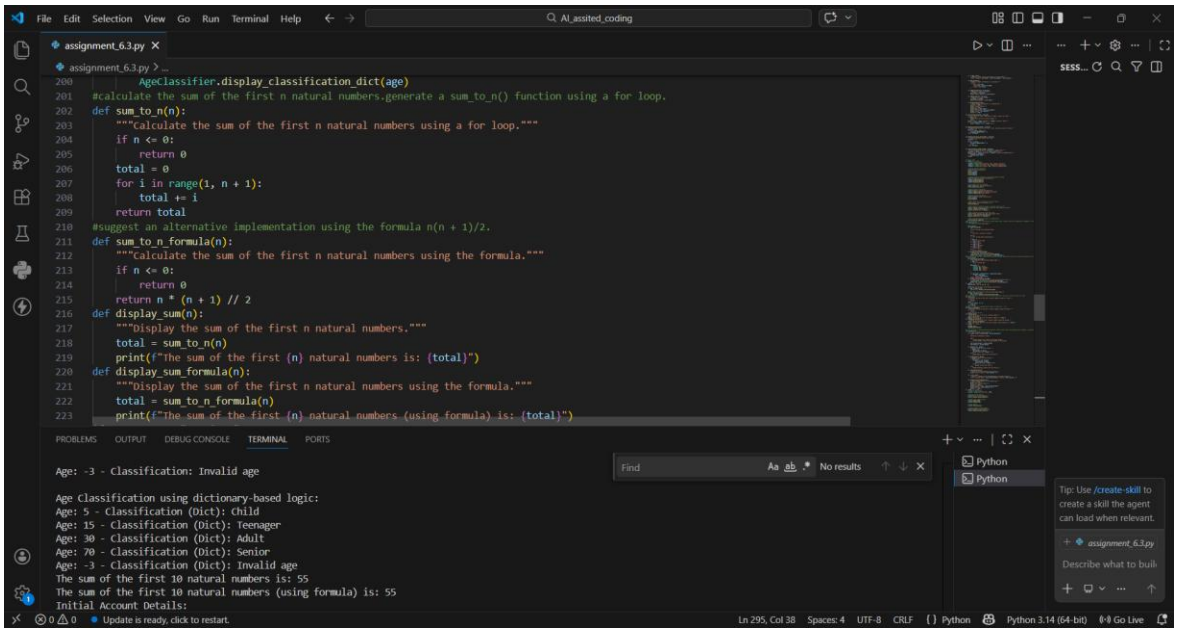


#generate generate the same classification using alternative conditional structures (simplified conditions or dictionary-based logic).

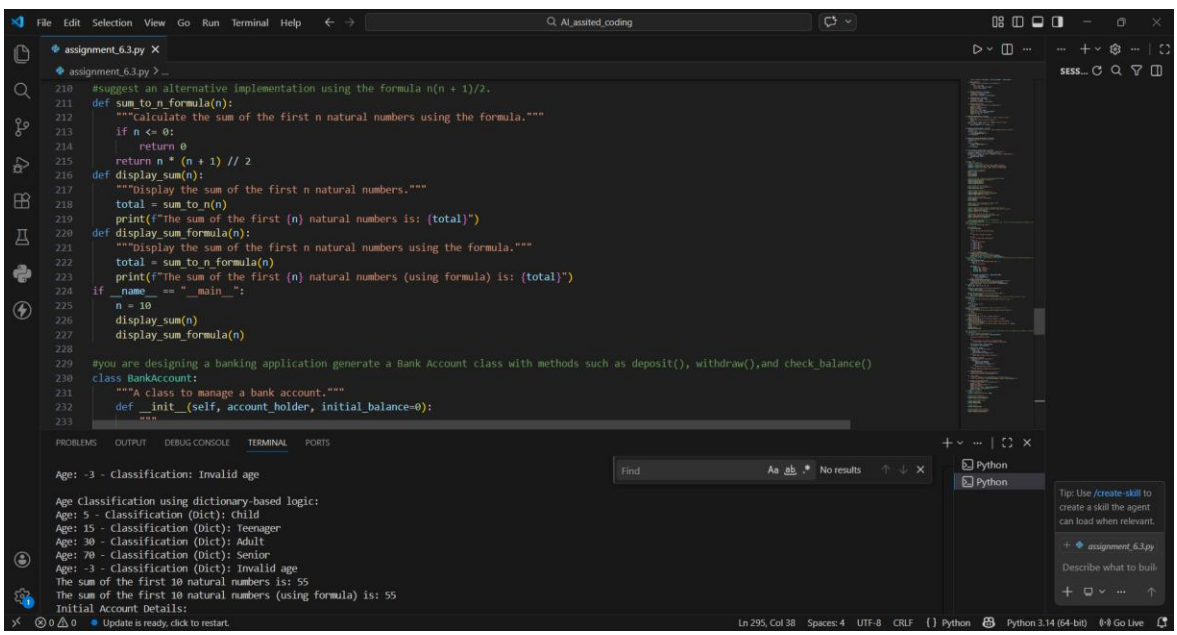


Task Description #4: For and While Loops (Sum of First n Numbers)

generate a sum_to_n() function using a for loop.



#suggest an alternative implementation using the formula n(n + 1)/2.



Task Description #5: Classes (Bank Account Class)

#you are designing a banking application generate a Bank Account class with methods such as deposit(), withdraw(),and check_balance()

```python
#you are designing a banking application generate a Bank Account class with methods such as deposit(), withdraw(),and check_balance()
class BankAccount:
    """A class to manage a bank account."""
    def __init__(self, account_holder, initial_balance=0):
        """
        Initialize a BankAccount object.

        Args:
            account_holder (str): Name of the account holder
            initial_balance (float): Initial balance of the account
        """
        self.account_holder = account_holder
        self.balance = initial_balance

    def deposit(self, amount):
        """Deposit money into the account."""
        if amount > 0:
            self.balance += amount
            print(f"Deposited: ${amount:.2f}")
        else:
            print("Deposit amount must be positive.")

    def withdraw(self, amount):
        """Withdraw money from the account."""
```

```
==================================================
Bank Account Details:
==================================================
Account Holder: John Doe
Balance: $1000.00
==================================================
Deposited: $500.00
Withdrew: $200.00
Current balance: $1300.00

Updated Account Details:
```

#Add meaningful comments and explain the working of the code

```python
class BankAccount:
    def display_account_details(self):
        """Print account details in a formatted way."""
        print(f"{'='*50}")
        print(f"Bank Account Details:")
        print(f"{'='*50}")
        print(f"Account Holder: {self.account_holder}")
        print(f"Balance: ${self.balance:.2f}")
        print(f"{'='*50}")
if __name__ == "__main__":
    # Create a bank account
    account = BankAccount("John Doe", 1000)

    # Display account details
    print("Initial Account Details:")
    account.display_account_details()

    # Deposit and withdraw money
    account.deposit(500)
    account.withdraw(200)

    # Check balance
    account.check_balance()

    # Display updated account details
    print("\nUpdated Account Details:")
    account.display_account_details()
```

```
==================================================
Account Holder: John Doe
Balance: $1000.00
==================================================
Deposited: $500.00
Withdrew: $200.00
Current balance: $1300.00
```